

Supplementary Online Materials to:

## Data and Code Availability in Sports Science: A Registered Report

Registered Report – Stage 1

PREPRINT VERSION – NOT PEER REVIEWED

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## 18 S1: Article Search Strings

19

20 Search string for Web of Science:

21

22 “(SO=(JOURNAL1 OR JOURNAL2 OR JOURNALK)) AND PY=(2017-2026) AND DT=(Article)”

23 where JOURNALK is a placeholder for the k journals that match the inclusion criteria.

24

## S2: Pilot Analysis

### Aim:

A pilot analysis was conducted to test the PDF retrieval and automated extraction tools. We randomly sampled 200 articles from a population sports science that will not be part of our study dataset. We documented the success rate and time required for automated and manual PDF retrieval and data extraction. We compared the results of the manual and automated extraction to determine the error rates of the automated procedure and to inform the extraction procedures for the main project.

### Methods:

We screened the SCImago Journal Rank 2025 (SCImago, 2025) in the subject category “Sports Science” for journals that were not listed as Q1 in the category during the previous ten years. We selected the ten journals with the highest 2025 rankings after excluding three journals that primarily publish non-quantitative research (International Review for the Sociology of Sport, Sociology of Sport Journal, Journal of the Philosophy of Sports). The remaining journals were: Journal of Human Kinetics; Strength and Conditioning Journal<sup>1</sup>; Journal of Sports Science and Medicine; Physical Therapy in Sport; Physician and Sportsmedicine; Sports Biomechanics; International Journal of Sports Science & Coaching; Journal of Sports Economics; Research Quarterly for Exercise and Sport; International Journal of Sports Marketing & Sponsorship.

We performed an article search using Web of Science on 28 August 2026 based on the inclusion criteria specified in our article. The exact search term was:

*“(SO=(Journal of Human Kinetics OR Strength and Conditioning Journal OR Journal of Sports Science and Medicine OR Physical Therapy in Sport OR Physician and Sportsmedicine OR Sports Biomechanics OR International Journal of Sports Science Coaching OR Journal of Sports Economics OR Research Quarterly for Exercise and Sport OR International Journal of Sports Marketing Sponsorship)) AND PY=(2017-2026) AND DT=(Article)”*

The search results were exported as .txt files and imported into R, where we took a random sample of n = 200 articles to constitute the pilot sample. We then attempted to retrieve the articles’ PDFs as outlined in the manuscript and documented the success rate for automated retrieval and the time required for manual retrieval.

To validate the automated screening process, we manually extracted all relevant items from the texts. We then conducted the automated screening and compared the results of the two approaches. During reviewing discrepancies between the automated and the manual extraction, we refined the automated extraction procedure and reran it.

### Results:

We could retrieve the full texts for n = 195 articles<sup>2</sup>. The Zotero import function for PDF retrieval worked for 53 (26.5%) articles. The average time required for retrieving a PDF manually was 16.5

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<sup>1</sup> During the reanalysis we realized that “Strength and Conditioning Journal” mainly publishes non-original research and thus does not meet the inclusion criteria. However, we retained the few articles from this journal (n = 9) in our sample to test whether the automated extraction can correctly identify that these articles do not contain data or code availability statements.

<sup>2</sup> Five articles were recently published in one journal that was included in our university’s subscription system only for articles assigned to an issue, but not for articles published ahead of print. Because this was a pilot study, we excluded these articles without resampling.

seconds. Manual extraction took an average of 52 seconds per study, while automated extraction took 6 seconds per article.

Data and/or code availability was automatically flagged in 42 cases, of which 29 corresponded to cases identified through manual extraction, 2 identified statements overlooked during manual extraction, and 13 were false positives (13/195, 7%). Note that these false positives would have been identified during the data extraction for the main project, which includes a manual review of each automatically flagged entry. The false negative rate was 0%, i.e., no data/code sharing or availability statement was overlooked by the automated procedure in the pilot sample.

Regarding the analysis software used, a total of 43 discrepancies across 40 articles were found. However, after correcting for errors (overlooked uses of software) in the manual extraction, only 17 discrepancies remained; 3 were false negatives (i.e., the automatic extraction overlooked the use of a software), and 14 were false positive (i.e., the automated extraction incorrectly detected the use of a software). Notably, 12 of the 14 false positives were related to the use of the software *R*, because regular expression matching proved difficult in these cases.

#### Conclusions:

- Retrieving PDFs is a bottleneck for scalability, as we can assume that 75% of the included studies will require manual retrieval, with approximately 16.5 seconds needed for each download.
- Automated extraction is approximately 10 times faster than manual extraction and has fewer extraction errors for the item analysis software used.
- Automatically flagging data/code availability statements has a false positive rate of around 7%, and thus requires manual oversight. The false negative rate is very low.
- Automatically detecting the analysis software in a study has false-positive and false-negative rates of approximately 1-2% each, provided that studies in which the software *R* is detected are manually reviewed (as these show high false positive rates).

### S3: Automated Extraction Strategy

Articles are flagged for manual extraction if one of the following conditions occur:

- A data or code availability statement is likely present:
  - o The grobid import recognizes an availability statements OR
  - o The full article text matches one of the following strings: "data availability" OR "code availability" OR "(on|by|upon)\s+(reasonable\s+)?request" OR "datasets?\s+(?:is|are)?\s\*(?:not\s+)?(?:publicly\s+)?available" OR "available\s+from\s+(?:the\s+)?(?:corresponding\s+)?author"
- Data or code is likely shared:
  - o The full article text matches one of the following strings that may indicate a data/code repository: "repositor" OR "archive" OR "\bosf\b" OR "\osf.io" OR "open science framework" OR "researchbox" OR "zenodo" OR "github" OR "figshare" OR "datadryad" OR "kaggle" or "mendeley" OR "ukbiobank" OR "ukdataservice" OR "dataverse" OR "clinicalstudydatarequest" OR "ourworldindata"

Some of the regular expression strings are modified from the code of the *open\_practices* module in the *metacheck* R package (DeBruine et al., 2026). Every article that is flagged will be manually reviewed.

For the determination of statistical software used, we use the following search strings:

- SPSS: "\bspss" OR "statistical package for the social sciences"
- Stata: "stata"
- jamovi: "jamovi"
- JASP: "\bjasp\b"
- R: "(?<!--)\bR(?:=\s(?:\s\*=))" (case-sensitive!) OR "(?:in|software)\s+R\b" (case-sensitive!) OR "a language and environment for statistical computing" OR "rstudio" OR "\br language"
- Python: "python" OR "scipy"
- GraphPad Prism: "graph pad" OR "\bprism\b"
- Excel: "excel\b"
- SAS: "\bsas\b"
- LISREL: "lisrel"
- MATLAB: "matlab"
- Statistica: "statistica"
- WinBUGS: "winbugs"
- BioEstat: "bioestat"
- Stan: "\bstan\b"
- PSPP: "\bpspp"
- MedCalc: "medcalc"

The list of software packages is in part informed by the work of Borg et al. (2020). When an article is flagged for using R, it will be manually reviewed.

The full R code for the above procedures is presented here:

```
#' Flag data availability statements and repository mention
#
#' Detects whether an article contains a data or code availability statement
#' and whether it mentions a data repository. Availability statements are
#' identified either through a dedicated availability section (identified by
#' the grobid import) or by searching
```

```

140 #' for common phrases (e.g., "data availability", "code availability", or
141 #' "available upon reasonable request"). Repository mentions are detected by
142 #' searching for references to common repositories such as OSF, Zenodo,
143 #' GitHub, Figshare, Dryad, Dataverse, and others.
144 #'
145 #' @param article An object with class "scivrs_paper", usually imported
146 #'   by the metacheck package
147 #'
148 #' @return A data frame with one row and two logical columns:
149 #' \describe{
150 #'   \item{flag_statement}{`TRUE` if the article contains a data/code
151 #'     availability statement or a dedicated availability section.}
152 #'   \item{flag_repo}{`TRUE` if the article mentions a supported data
153 #'     repository.}
154 #' }
155 #'
156 flag_article <- function(article) {
157   # search for availability section in grobid import
158   if (is.null(article$section)) article <- article[[1]] # get list item if
159   needed
160   sec <- any(article$section$section_type == "availability")
161   # search for strings that indicate availability section
162   st <- sfind(
163     article,
164     c("data availability", "code availability",
165       "(on|by|upon)\\s+(reasonable\\s+)?request",
166       "datasets?\\s+(?:is|are)?\\s*(?:not\\s+)?(?:publicly\\s+)?available",
167       "available\\s+from\\s+(?:the\\s+)?(?:corresponding\\s+)?author"
168     ),
169     search_header = TRUE
170   )
171   # search for strings that match data/code repository
172   repo <- sfind(
173     article,
174     c("repositor", "archive", "\\bosf\\b", "osf\\.io",
175       "open science framework", "researchbox", "zenodo", "github",
176       "figshare", "datadryad", "kaggle", "mendeley", "ukbiobank",
177       "ukdataservice", "dataverse", "clinicalstudydatarequest",
178       "ourworldindata")
179   )
180   data.frame(
181     flag_statement = any(c(sec, st)),
182     flag_repo = repo
183   )
184 }
185
186 #' Detect statistical software mentioned in an article
187 #'
188 #' Searches an article for mentions of commonly used statistical software and

```

```

189 #' programming languages. Detection is based on regular expressions and common
190 #' alternative names where appropriate (e.g., "Statistical Package for the
191 #' Social Sciences" for SPSS or "RStudio" for R).
192 #'
193 #' @param article An object with class "scivrs_paper", usually imported
194 #'   by the metacheck package
195 #'
196 #' @return A data frame with one row containing logical columns indicating
197 #'   whether each software package was detected. The returned columns are:
198 #'   `spss`, `stata`, `jamovi`, `jasp`, `r`, `python`, `prism`, `excel`,
199 #'   `sas`, `lisrel`, `matlab`, `statistica`, `winbugs`, `bioestat`,
200 #'   `stan`, `pspp`, and `medcalc`.
201 #'
202 extract_software <- function(article) {
203   data.frame(
204     spss = sfind(article, c("\\bspss", "statistical package for the social
205 sciences", "\\bpasw", "\\bamos")),
206     stata = sfind(article, "\\bstata"),
207     jamovi = sfind(article, "\\bjamovi"),
208     jasp = sfind(article, "\\bjasp"),
209     r = any(
210       sfind(article, c("(?!-)\bR(?:=\\s(?:!\\s*))",
211 "(?:in|software)\\s+R\\b")), ignore.case = FALSE, perl = TRUE),
212       sfind(article, c("a language and environment for statistical
213 computing", "rstudio", "\\br language")))
214   ),
215   python = sfind(article, c("python", "scipy")),
216   prism = sfind(article, c("graph pad", "\\bprism\\b")),
217   excel = sfind(article, "excel\\b"),
218   sas = sfind(article, "\\bsas\\b"),
219   lisrel = sfind(article, "lisrel"),
220   matlab = sfind(article, c("matlab", "mathworks")),
221   statistica = sfind(article, "\\bstatistica\\b"),
222   winbugs = sfind(article, "winbugs"),
223   bioestat = sfind(article, "bioestat"),
224   stan = sfind(article, "\\bstan\\b"),
225   pspp = sfind(article, "\\bpspp"),
226   medcalc = sfind(article, "medcalc")
227 )
228 }
229 #' Helper function to find matches using metacheck::text_search()
230 #'
231 #' Searches an article text via metacheck::text_search() and returns
232 #' if any matches were found
233 #'
234 #' @param article An object with class "scivrs_paper", usually imported
235 #'   by the metacheck package
236 #' @param string Regular expression pattern passed to `pattern` in
237 #' metacheck::text_search()

```

```
238 #' @param ... Additional arguments passed to metacheck::text_search()
239 #'
240 #' @return TRUE or FALSE, whether a match was found.
241 sfind <- function(article, string, ...) {
242   t <- metacheck::text_search(paper = article, pattern = string, ...)
243   nrow(t) >= 1
244 }
245
```

## References

- Borg, D. N., Bon, J. J., Sainani, K. L., Baguley, B. J., Tierney, N. J., & Drovandi, C. (2020). Comment on: 'Moving Sport and Exercise Science Forward: A Call for the Adoption of More Transparent Research Practices.' *Sports Medicine*, 50(8), 1551–1553.  
<https://doi.org/10.1007/s40279-020-01298-5>
- DeBruine, L., Mesquida, C., Werner, J., & Lakens, D. (2026). *metacheck: Check research outputs for best practices* [Manual]. <https://doi.org/10.5281/zenodo.20704754>
- SCImago. (2025). *SJR-SCImago Journal Country & Rank [Portal]*.  
<https://www.scimagojr.com/journalrank.php?category=3699&year=2025>